

TRIGONOMETRY

- $\pi^c = 180^0$, $1^0 = 60'$, $1' = 60''$, $1^0 = 0.01745 \text{ radians}$ and $1^c = 57^0 17' 45''$.

- **Trigonometric Ratios:**

$$(1) \quad \sin \theta = \frac{1}{\operatorname{cosec} \theta} = \frac{p}{h}.$$

$$(2) \quad \cos \theta = \frac{1}{\sec \theta} = \frac{b}{h}.$$

$$(3) \quad \tan \theta = \frac{1}{\cot \theta} = \frac{p}{b}.$$

$$(4) \quad \cot \theta = \frac{1}{\tan \theta} = \frac{b}{p}.$$

$$(5) \quad \sec \theta = \frac{1}{\cos \theta} = \frac{h}{b}.$$

$$(6) \quad \operatorname{cosec} \theta = \frac{1}{\sin \theta} = \frac{h}{p}.$$

- **Trigonometric Relations:**

$$(1) \quad \sin \theta \times \operatorname{cosec} \theta = 1.$$

$$(2) \quad \cos \theta \times \sec \theta = 1.$$

$$(3) \quad \tan \theta \times \cot \theta = 1.$$

$$(4) \quad \sin^2 \theta + \cos^2 \theta = 1, \quad \sin^2 \theta = 1 - \cos^2 \theta, \quad \cos^2 \theta = 1 - \sin^2 \theta.$$

$$(5) \quad \sec^2 \theta - \tan^2 \theta = 1, \quad \sec^2 \theta = 1 + \tan^2 \theta, \quad \tan^2 \theta = \sec^2 \theta - 1.$$

$$(6) \quad \operatorname{cosec}^2 \theta - \cot^2 \theta = 1, \quad \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta, \quad \cot^2 \theta = \operatorname{cosec}^2 \theta - 1.$$

$$(7) \quad \sec^2 \theta + \operatorname{cosec}^2 \theta = \sec^2 \theta \times \operatorname{cosec}^2 \theta.$$

$$(8) \quad \tan^2 \theta - \sin^2 \theta = \tan^2 \theta \times \sin^2 \theta.$$

$$(9) \quad \cot^2 \theta - \cos^2 \theta = \cot^2 \theta \times \cos^2 \theta.$$

$$(10) \quad \sin^2 \theta + \cos^4 \theta = 1 - \sin^2 \theta \cos^2 \theta = \sin^4 \theta + \cos^2 \theta.$$

$$(11) \quad \sin^4 \theta + \cos^4 \theta = 1 - 2\sin^2 \theta \cos^2 \theta.$$

$$(12) \quad \sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta.$$

$$(13) \quad \sin^2 \theta + \operatorname{cosec}^2 \theta \geq 2.$$

$$(14) \quad \cos^2 \theta + \sec^2 \theta \geq 2.$$

$$(15) \quad \tan^2 \theta + \cot^2 \theta \geq 2.$$

• Values of Trigonometric Ratios:

Angle Ratio	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{1}{\sqrt{2}}$	$-\frac{\sqrt{3}}{2}$	-1
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	-1	$-\frac{1}{\sqrt{3}}$	
cot	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	$-\frac{1}{\sqrt{3}}$	-1	$-\sqrt{3}$	∞
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	-2	$-\sqrt{2}$	$-\frac{2}{\sqrt{3}}$	-1
cos ec	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞

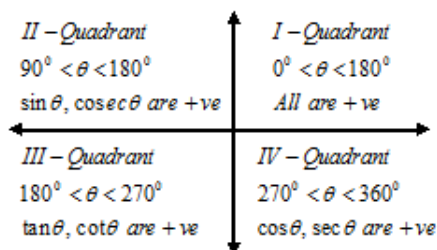
θ	15°	$22\frac{1}{2}^\circ$	$67\frac{1}{2}^\circ$	75°
sin	$\frac{\sqrt{3}-1}{2\sqrt{2}}$	$\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$	$\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$	$\frac{\sqrt{3}+1}{2\sqrt{2}}$
cos	$\frac{\sqrt{3}+1}{2\sqrt{2}}$	$\sqrt{\frac{\sqrt{2}+1}{2\sqrt{2}}}$	$\sqrt{\frac{\sqrt{2}-1}{2\sqrt{2}}}$	$\frac{\sqrt{3}-1}{2\sqrt{2}}$
tan	$2-\sqrt{3}$	$\sqrt{2}-1$	$\sqrt{2}+1$	$2+\sqrt{3}$
cot	$2+\sqrt{3}$	$\sqrt{2}+1$	$\sqrt{2}-1$	$2-\sqrt{3}$
sec	$\sqrt{6}-\sqrt{2}$	$\sqrt{4-2\sqrt{2}}$	$\sqrt{4+2\sqrt{2}}$	$\sqrt{6}+\sqrt{2}$
cosec	$\sqrt{6}+\sqrt{2}$	$\sqrt{4+2\sqrt{2}}$	$\sqrt{4-2\sqrt{2}}$	$\sqrt{6}-\sqrt{2}$

θ	18°	36°	54°	72°
sin	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10+2\sqrt{5}}}{4}$
cos	$\frac{\sqrt{10+2\sqrt{5}}}{4}$	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{10-2\sqrt{5}}}{4}$	$\frac{\sqrt{5}-1}{4}$

• Domain and Range of Trigonometric functions:

Function	Domain	Range
$\sin \theta$	R	$[-1,1]$
$\cos \theta$	R	$[-1,1]$
$\tan \theta$	$R - \left\{ (2n+1)\frac{\pi}{2}, n \in Z \right\}$	R
$\cot \theta$	$R - \{n\pi, n \in Z\}$	R
$\sec \theta$	$R - \left\{ (2n+1)\frac{\pi}{2}, n \in Z \right\}$	$(-\infty, -1] \cup [1, \infty)$
$\operatorname{cosec} \theta$	$R - \{n\pi, n \in Z\}$	$(-\infty, -1] \cup [1, \infty)$
$\sin^{-1} \theta$	$[-1,1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$
$\cos^{-1} \theta$	$[-1,1]$	$[0, \pi]$
$\tan^{-1} \theta$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$
$\cot^{-1} \theta$	$(-\infty, \infty)$	$(0, \pi)$
$\sec^{-1} \theta$	$(-\infty, -1] \cup [1, \infty)$	$\left(0, \frac{\pi}{2} \right) \cup \left(\frac{\pi}{2}, \pi \right)$
$\operatorname{cosec}^{-1} \theta$	$(-\infty, -1] \cup [1, \infty)$	$\left(-\frac{\pi}{2}, 0 \right) \cup \left(0, \frac{\pi}{2} \right)$
$\sinh \theta$	R	R
$\cosh \theta$	R	$[1, \infty)$
$\tanh \theta$	R	$(-1, 1)$
$\coth \theta$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, -1) \cup (1, \infty)$
$\operatorname{sech} \theta$	R	$(0, 1]$
$\operatorname{cosech} \theta$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$\sinh^{-1} \theta$	R	R
$\cosh^{-1} \theta$	$[1, \infty)$	$[0, \infty)$
$\tanh^{-1} \theta$	$(-1, 1)$	R
$\coth^{-1} \theta$	$(-\infty, -1) \cup (1, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$\operatorname{sech}^{-1} \theta$	$(0, 1]$	$[0, \infty)$
$\operatorname{cosech}^{-1} \theta$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$

• **Sign of Trigonometric Ratios (ASTC Rule):**



- (1) $0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ, 450^\circ, \dots$ are called as quadrant angles.
- (2) Co-terminal Angles: If two angles differ by an integral multiple of 360° then two angles are called as co-terminal angles. $30^\circ, 390^\circ, 750^\circ, -330^\circ, \dots$ are co-terminal angles.

• **Increasing and Decreasing behavior of trigonometric ratios:**

- (1) In Quadrant-I: $\sin \theta, \tan \theta, \sec \theta$ are increasing functions and $\cos \theta, \cot \theta, \operatorname{cosec} \theta$ are decreasing functions.
- (2) In Quadrant-II: $\tan \theta, \sec \theta, \operatorname{cosec} \theta$ are increasing functions and $\sin \theta, \cos \theta, \cot \theta$ are decreasing functions.
- (3) In Quadrant-III: $\tan \theta, \cos \theta, \operatorname{cosec} \theta$ are increasing functions and $\sin \theta, \cot \theta, \sec \theta$ are decreasing functions.
- (4) In Quadrant-IV: $\sin \theta, \cos \theta, \tan \theta$ are increasing functions and $\sec \theta, \cot \theta, \operatorname{cosec} \theta$ are decreasing functions.

• **Trigonometric Ratios of different angles:**

	$90^\circ - \theta$	$90^\circ + \theta$	$180^\circ - \theta$	$180^\circ + \theta$	$270^\circ - \theta$	$270^\circ + \theta$	$360^\circ - \theta$	$360^\circ + \theta$
sin	$\cos \theta$	$\cos \theta$	$\sin \theta$	$-\sin \theta$	$-\cos \theta$	$-\cos \theta$	$-\sin \theta$	$\sin \theta$
cos	$\sin \theta$	$-\sin \theta$	$-\cos \theta$	$-\cos \theta$	$-\sin \theta$	$\sin \theta$	$\cos \theta$	$\cos \theta$
tan	$\cot \theta$	$-\cot \theta$	$-\tan \theta$	$\tan \theta$	$\cot \theta$	$-\cot \theta$	$-\tan \theta$	$\tan \theta$
cot	$\tan \theta$	$-\tan \theta$	$-\cot \theta$	$\cot \theta$	$\tan \theta$	$-\tan \theta$	$-\cot \theta$	$\cot \theta$
sec	$\operatorname{cosec} \theta$	$-\operatorname{cosec} \theta$	$-\sec \theta$	$-\sec \theta$	$-\operatorname{cosec} \theta$	$\operatorname{cosec} \theta$	$\sec \theta$	$\sec \theta$
cosec	$\sec \theta$	$\sec \theta$	$\operatorname{cosec} \theta$	$-\operatorname{cosec} \theta$	$-\sec \theta$	$-\sec \theta$	$-\operatorname{cosec} \theta$	$\operatorname{cosec} \theta$

• Formula:

- (1) $\sin(n \times 360^\circ + \theta) = \sin \theta$.
- (2) $\cos(n \times 360^\circ + \theta) = \cos \theta$.
- (3) $\tan(n \times 360^\circ + \theta) = \tan \theta$.
- (4) $\cot(n \times 360^\circ + \theta) = \cot \theta$.
- (5) $\sec(n \times 360^\circ + \theta) = \sec \theta$.
- (6) $\operatorname{cosec}(n \times 360^\circ + \theta) = \operatorname{cosec} \theta$.

• Formula:

- (1) $\sin(n \times 360^\circ - \theta) = -\sin \theta$.
- (2) $\cos(n \times 360^\circ - \theta) = \cos \theta$.
- (3) $\tan(n \times 360^\circ - \theta) = -\tan \theta$.
- (4) $\cot(n \times 360^\circ - \theta) = -\cot \theta$.
- (5) $\sec(n \times 360^\circ - \theta) = \sec \theta$.
- (6) $\operatorname{cosec}(n \times 360^\circ - \theta) = -\operatorname{cosec} \theta$.

• If two angles A and B are complementary is $A + B = 90^\circ$, then:

- (1) $\sin A = \cos B$.
- (2) $\cos A = \sin B$.
- (3) $\sin^2 A + \sin^2 B = 1$.
- (4) $\cos^2 A + \cos^2 B = 1$.
- (5) $\tan A \tan B = 1$.
- (6) $\cot A \cot B = 1$.

• If two angles A and B are supplementary is $A + B = 180^\circ$, then:

- (1) $\sin A - \sin B = 0$.
- (2) $\cos A + \cos B = 0$.
- (3) $\tan A + \tan B = 0$.

• If $A - B = 180^\circ$, then:

- (1) $\sin A + \sin B = 0$.
- (2) $\cos A + \cos B = 0$.
- (3) $\tan A - \tan B = 0$.

• If $A + B = 360^\circ$, then:

- (1) $\sin A + \sin B = 0$.
- (2) $\cos A - \cos B = 0$.
- (3) $\tan A + \tan B = 0$.

• If $a \cos \theta + b \sin \theta = c$, then $a \sin \theta - b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}$.

• If $a \cos \theta - b \sin \theta = c$, then $a \sin \theta + b \cos \theta = \pm \sqrt{a^2 + b^2 - c^2}$.

$$\bullet \quad \sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + (n-1)\beta) = \frac{\sin \left\{ \alpha + (n-1) \frac{\beta}{2} \right\} \sin \left(\frac{n\beta}{2} \right)}{\sin \left(\frac{\beta}{2} \right)}.$$

$$\bullet \quad \cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \dots + \cos(\alpha + (n-1)\beta) = \frac{\cos \left\{ \alpha + (n-1) \frac{\beta}{2} \right\} \sin \left(\frac{n\beta}{2} \right)}{\sin \left(\frac{\beta}{2} \right)}$$

- $\sin \theta + \sin(\pi + \theta) + \sin(2\pi + \theta) + \sin(3\pi + \theta) + \dots + \sin(n\pi + \theta) = \begin{cases} 0 & \text{if } n \text{ odd} \\ \sin \theta & \text{if } n \text{ even} \end{cases}$
- $\cos \theta + \cos(\pi + \theta) + \cos(2\pi + \theta) + \cos(3\pi + \theta) + \dots + \cos(n\pi + \theta) = \begin{cases} 0 & \text{if } n \text{ odd} \\ \cos \theta & \text{if } n \text{ even} \end{cases}$
- **Compound Angles:** The algebraic sum of two or more angles is called as compound angles i.e. $A+B$, $A+C$, $B+C$, $A-B$, $C-A$, $B-C$, $A+B+C$, $A+B-C$, $A-B+C$,...
- (1) $\sin(A+B) = \sin A \cos B + \cos A \sin B$.
- (2) $\sin(A-B) = \sin A \cos B - \cos A \sin B$.
- (3) $\cos(A+B) = \cos A \cos B - \sin A \sin B$.
- (4) $\cos(A-B) = \cos A \cos B + \sin A \sin B$.
- (5) $\sin(A+B) + \sin(A-B) = 2 \sin A \cos B$.
- (6) $\sin(A+B) - \sin(A-B) = 2 \cos A \sin B$.
- (7) $\cos(A-B) + \cos(A+B) = 2 \cos A \cos B$.
- (8) $\cos(A-B) - \cos(A+B) = 2 \sin A \sin B$.
- (9) $\cos(A+B) - \cos(A-B) = -2 \sin A \sin B$.
- (10) $\sin(A+B) \times \sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$.
- (11) $\cos(A+B) \times \cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$.
- (12) $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.
- (13) $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$.
- (14) $\cot(A+B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}$.
- (15) $\cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$.
- (16) $\tan A + \tan B = \frac{\sin(A+B)}{\cos A \cos B}$.
- (17) $\tan A - \tan B = \frac{\sin(A-B)}{\cos A \cos B}$.
- (18) $\cot A + \cot B = \frac{\cos(A-B)}{\sin A \sin B}$.
- (19) $\cot A - \cot B = \frac{-\cos(A+B)}{\sin A \sin B}$.
- (20) $\tan(A+B) \times \tan(A-B) = \frac{\tan^2 A - \tan^2 B}{1 - \tan^2 A \tan^2 B}$.
- (21) $\cot(A+B) \times \cot(A-B) = \frac{\cot^2 A \cot^2 B - 1}{\cot^2 B - \cot^2 A}$.
- (22) $\tan(45^\circ + \theta) = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} = \cot(45^\circ - \theta) = \frac{1 + \tan \theta}{1 - \tan \theta} = \frac{\cot \theta + 1}{\cot \theta - 1}$.

- (23) $\tan(45^\circ - \theta) = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} = \cot(45^\circ + \theta) = \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{\cot \theta - 1}{\cot \theta + 1}.$
- (24) $\tan(45^\circ + \theta) \times \tan(45^\circ - \theta) = 1.$
- (25) If $\tan \alpha = \frac{m}{m+1}$; $\tan \beta = \frac{1}{2m+1}$ then $\tan(\alpha + \beta) = 1.$
- (26) If $\tan \alpha = \frac{m+1}{m}$; $\tan \beta = \frac{1}{2m+1}$ then $\tan(\alpha - \beta) = 1.$
- (27) If $\tan \alpha = \frac{m}{m-1}$; $\tan \beta = \frac{1}{2m-1}$ then $\tan(\alpha - \beta) = 1.$
- (28) If $\tan \alpha = \frac{m}{n}$; $\tan \beta = \frac{n-m}{n+m}$ then $\tan(\alpha + \beta) = 1.$
- (29) If $\tan \alpha = \frac{m}{n}$; $\tan \beta = \frac{m-n}{m+n}$ then $\tan(\alpha - \beta) = 1.$
- (30) $\tan A + \tan B = \frac{\sin(A+B)}{\cos A \cos B}.$
- (31) $\tan A - \tan B = \frac{\sin(A-B)}{\cos A \cos B}.$
- (32) $\cot A + \cot B = \frac{\sin(B+A)}{\sin A \sin B}.$
- (33) $\cot A - \cot B = \frac{\sin(B-A)}{\sin A \sin B}.$
- (34) $\sin \theta - \sin(120^\circ - \theta) + \sin(120^\circ - \theta) = 0.$
- (35) $\sin \theta - \sin(240^\circ - \theta) + \sin(240^\circ - \theta) = 0.$
- (36) $\cos \theta + \cos(120^\circ - \theta) + \cos(120^\circ - \theta) = 0.$
- (37) $\cos \theta + \cos(240^\circ - \theta) + \cos(240^\circ - \theta) = 0.$
- (38) If $A + C = B$, then $\tan B - \tan A - \tan C = \tan A \tan B \tan C.$
- If $A + B = 45^\circ$ or $A + B = 225^\circ$, then:
 - (1) $(1 + \tan A)(1 + \tan B) = 2.$
 - (2) $(1 + \cot A)(1 + \cot B) = 2 \cot A \cot B.$
 - (3) $(1 - \cot A)(1 - \cot B) = 2.$
 - If $A + B = 135^\circ$ or $A + B = 315^\circ$, then:
 - (1) $(1 - \tan A)(1 - \tan B) = 2.$
 - (2) $(1 + \tan A)(1 + \tan B) = 2 \tan A \tan B.$
 - (3) $(1 + \cot A)(1 + \cot B) = 2.$
 - $\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C.$
 - $\cos(A + B + C) = \cos A \cos B \cos C - \sin A \sin B \cos C - \sin A \cos B \sin C - \cos A \sin B \sin C.$

- $\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$.
- $\cot(A + B + C) = \frac{\cot A + \cot B + \cot C - \cot A \cot B \cot C}{1 - \cot A \cot B - \cot B \cot C - \cot C \cot A}$.

• **In General:**

(1) $\sin(A_1 + A_2 + \dots + A_n) = \cos A_1 \cos A_2 \dots \cos A_n (S_1 - S_3 + S_5 - S_7 + \dots)$.

(2) $\cos(A_1 + A_2 + \dots + A_n) = \cos A_1 \cos A_2 \dots \cos A_n (1 - S_2 + S_4 - S_6 + S_8 + \dots)$.

(3) $\tan(A_1 + A_2 + \dots + A_n) = \frac{S_1 - S_3 + S_5 - S_7 + \dots}{1 - S_2 + S_4 - S_6 + S_8 + \dots}$.

(4) Where :

$S_1 = \tan A_1 + \tan A_2 + \dots + \tan A_n =$ Sum of the tangents of all the angles.

$S_2 = \tan A_1 \tan A_2 + \tan A_1 \tan A_3 + \dots =$ Sum of tangents taken two at a time.

$S_3 = \tan A_1 \tan A_2 \tan A_3 + \tan A_1 \tan A_2 \tan A_4 + \dots =$ Sum of tangents taken three at a time.

$S_n = \tan A_1 \tan A_2 \dots \tan A_n =$ Product of all the tangents.

(5) If $A_1 = A_2 = \dots = A_n$, then

$S_1 = n \tan A_1$.

$S_2 = {}^nC_2 \tan^2 A_1$.

$S_3 = {}^nC_3 \tan^3 A_1$.

$S_r = {}^nC_r \tan^r A_1$.

$S_n = \tan^n A_1$.

(6) $\sin nA = \cos^n A ({}^nC_1 \tan A - {}^nC_3 \tan^3 A + {}^nC_5 \tan^5 A - {}^nC_7 \tan^7 A + \dots)$.

(7) $\cos nA = \cos^n A (1 - {}^nC_2 \tan^2 A + {}^nC_4 \tan^4 A - {}^nC_6 \tan^6 A + {}^nC_8 \tan^8 A + \dots)$.

(8) $\tan nA = \frac{{}^nC_1 \tan A - {}^nC_3 \tan^3 A + {}^nC_5 \tan^5 A - {}^nC_7 \tan^7 A + \dots}{1 - {}^nC_2 \tan^2 A + {}^nC_4 \tan^4 A - {}^nC_6 \tan^6 A + {}^nC_8 \tan^8 A + \dots}$.

- If $A + B + C = (2n+1)\frac{\pi}{2}$, then:

(1) $\cot A + \cot B + \cot C = \cot A \cot B \cot C$.

(2) $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$.

- If $A + B + C = n\pi$, then:

(1) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

(2) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$.

• **Multiple Angle:**

- (1) $\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}.$
- (2) $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2} = \frac{2 \tan \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}.$
- (3) $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}.$
- (4) $\cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = 2 \cos^2 \frac{A}{2} - 1 = 1 - 2 \sin^2 \frac{A}{2} = \frac{1 - \tan^2 \frac{A}{2}}{1 + \tan^2 \frac{A}{2}}.$
- (5) $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}.$
- (6) $\tan A = \frac{2 \tan \frac{A}{2}}{1 - \tan^2 \frac{A}{2}}.$
- (7) $\cot 2A = \frac{\cot^2 A - 1}{2 \cot A}.$
- (8) $\cot A = \frac{\cot^2 \frac{A}{2} - 1}{2 \cot \frac{A}{2}}.$
- (9) $\tan\left(\frac{\pi}{4} + \frac{A}{2}\right) = \frac{1 + \tan \frac{A}{2}}{1 - \tan \frac{A}{2}} = \frac{\cos \frac{A}{2} + \sin \frac{A}{2}}{\cos \frac{A}{2} - \sin \frac{A}{2}} = \sqrt{\frac{1 + \sin A}{1 - \sin A}} = \frac{1 + \sin A}{\cos A} = \frac{\cos A}{1 - \sin A} = \sec A + \tan A = \cot\left(\frac{\pi}{4} - \frac{A}{2}\right).$
- (10) $\tan\left(\frac{\pi}{4} - \frac{A}{2}\right) = \frac{1 - \tan \frac{A}{2}}{1 + \tan \frac{A}{2}} = \frac{\cos \frac{A}{2} - \sin \frac{A}{2}}{\cos \frac{A}{2} + \sin \frac{A}{2}} = \sqrt{\frac{1 - \sin A}{1 + \sin A}} = \frac{1 - \sin A}{\cos A} = \frac{\cos A}{1 + \sin A} = \sec A - \tan A = \cot\left(\frac{\pi}{4} + \frac{A}{2}\right).$
- (11) $\sin 3A = 3 \sin A - 4 \sin^3 A = 4 \sin(60^\circ - A) \cdot \sin A \cdot \sin(60^\circ + A).$
- (12) $\sin^3 A = \frac{\sin 3A - 3 \sin A}{4}.$
- (13) $\cos 3A = 4 \cos^3 A - 3 \cos A = 4 \cos(60^\circ - A) \cdot \cos A \cdot \cos(60^\circ + A).$
- (14) $\cos^3 A = \frac{\cos 3A + 3 \cos A}{4}.$
- (15) $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A} = \tan(60^\circ - A) \cdot \tan A \cdot \tan(60^\circ + A).$
- (16) $\sin 4A = 4 \sin A \cos^3 A - 4 \cos A \sin^3 A.$
- (17) $\cos 4A = 8 \cos^4 A - 8 \cos^2 A + 1.$
- (18) $\tan 4A = \frac{4 \tan A - 4 \tan^3 A}{1 - 6 \tan^2 A + \tan^4 A}.$
- (19) $\sin 5A = 16 \sin^5 A - 20 \sin^3 A + 5 \sin A.$
- (20) $\cos 5A = 16 \cos^5 A - 20 \cos^3 A + 5 \cos A.$
- (21) $\sin^2 A = \frac{1 - \cos 2A}{2}; \sin A = \pm \sqrt{\frac{1 - \cos 2A}{2}}.$
- (22) $\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}; \sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}.$

- (23) $\cos^2 A = \frac{1+\cos 2A}{2}$; $\cos A = \pm \sqrt{\frac{1+\cos 2A}{2}}$.
- (24) $\cos^2 \frac{A}{2} = \frac{1+\cos A}{2}$; $\cos \frac{A}{2} = \pm \sqrt{\frac{1+\cos A}{2}}$.
- (25) $\tan^2 A = \frac{1-\cos 2A}{1+\cos 2A}$; $\tan A = \pm \sqrt{\frac{1-\cos 2A}{1+\cos 2A}}$.
- (26) $\tan^2 \frac{A}{2} = \frac{1-\cos A}{1+\cos A}$; $\tan \frac{A}{2} = \pm \sqrt{\frac{1-\cos A}{1+\cos A}}$.
- (27) $1 + \sin 2A = (\cos A + \sin A)^2$.
- (28) $1 - \sin 2A = (\cos A - \sin A)^2$.
- (29) $1 + \sin A = \left(\cos \frac{A}{2} + \sin \frac{A}{2}\right)^2$.
- (30) $1 - \sin A = \left(\cos \frac{A}{2} - \sin \frac{A}{2}\right)^2$.
- (31) $\cos A + \sin A = \pm \sqrt{1 + \sin 2A}$.
- (32) $\cos A - \sin A = \pm \sqrt{1 - \sin 2A}$.
- (33) $\cos \frac{A}{2} + \sin \frac{A}{2} = \pm \sqrt{1 + \sin A}$.
- (34) $\cos \frac{A}{2} - \sin \frac{A}{2} = \pm \sqrt{1 - \sin A}$.
- (35) $2 \sin A = \pm \sqrt{1 + \sin 2A} \pm \sqrt{1 - \sin 2A}$.
- (36) $2 \sin \frac{A}{2} = \pm \sqrt{1 + \sin A} \pm \sqrt{1 - \sin A}$.
- (37) $2 \cos A = \pm \sqrt{1 + \sin 2A} \pm \sqrt{1 - \sin 2A}$.
- (38) $2 \cos \frac{A}{2} = \pm \sqrt{1 + \sin A} \pm \sqrt{1 - \sin A}$.
- (39) $\sin \frac{A}{2} + \cos \frac{A}{2} > 0$; $\sin \frac{A}{2} - \cos \frac{A}{2} > 0$ if $\frac{\pi}{4} < \frac{A}{2} < \frac{3\pi}{4}$.
- (40) $\sin \frac{A}{2} + \cos \frac{A}{2} < 0$; $\sin \frac{A}{2} - \cos \frac{A}{2} > 0$ if $\frac{3\pi}{4} < \frac{A}{2} < \frac{5\pi}{4}$.
- (41) $\sin \frac{A}{2} + \cos \frac{A}{2} < 0$; $\sin \frac{A}{2} - \cos \frac{A}{2} < 0$ if $\frac{5\pi}{4} < \frac{A}{2} < \frac{7\pi}{4}$.
- (42) $\sin \frac{A}{2} + \cos \frac{A}{2} > 0$; $\sin \frac{A}{2} - \cos \frac{A}{2} < 0$ if $-\frac{\pi}{4} < \frac{A}{2} < \frac{\pi}{4}$.
- (43) $\cot A + \tan A = 2 \operatorname{cosec} 2A$.
- (44) $\cot A - \tan A = 2 \cot 2A$.
- (45) $\tan\left(\frac{\pi}{4} + A\right) + \tan\left(\frac{\pi}{4} - A\right) = 2 \sec 2A$.
- (46) $\tan\left(\frac{\pi}{4} + A\right) - \tan\left(\frac{\pi}{4} - A\right) = 2 \tan 2A$.
- (47) $\tan A + 2 \tan 2A + 2^2 \tan 2^2 A + 2^3 \tan 2^3 A + \dots + 2^n \tan 2^n A = \cot A$.
- (48) $\sin \theta \times \sin(\alpha - \theta) \times \sin(\alpha + \theta) = \frac{\sin 3\theta}{4}$, where $\alpha = 60^\circ$ or 120° or 240° or 300° .
- (49) $\cos \theta \times \cos(\alpha - \theta) \times \cos(\alpha + \theta) = \frac{\cos 3\theta}{4}$, where $\alpha = 60^\circ$ or 120° or 240° or 300° .
- (50) $\tan \theta \times \tan(\alpha - \theta) \times \tan(\alpha + \theta) = \tan 3\theta$, where $\alpha = 60^\circ$ or 120° or 240° or 300° .
- (51) $\cot \theta \times \cot(\alpha - \theta) \times \cot(\alpha + \theta) = \cot 3\theta$, where $\alpha = 60^\circ$ or 120° or 240° or 300° .

- (52) $\cos \theta \times \cos 2\theta \times \cos 2^2 \theta \times \dots \times \cos 2^{n-1} \theta = \frac{\sin 2^n \theta}{2^n \sin \theta}.$
- (53) If $\theta = \frac{\pi}{2^n + 1}$, then $\cos \theta \times \cos 2\theta \times \cos 2^2 \theta \times \dots \times \cos 2^{n-1} \theta = \frac{1}{2^n}.$
- (54) If $\theta = \frac{\pi}{2^n - 1}$, then $\cos \theta \times \cos 2\theta \times \cos 2^2 \theta \times \dots \times \cos 2^{n-1} \theta = -\frac{1}{2^n}.$
- (55) $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos n\theta = \frac{\sin \frac{n\theta}{2} \cos \frac{(n+1)\theta}{2}}{\sin \frac{\theta}{2}}.$
- (56) $\sin^2 \theta + \sin^2 (\alpha - \theta) + \sin^2 (\alpha + \theta) = \frac{3}{2}.$
- (57) $\cos^2 \theta + \cos^2 (\alpha - \theta) + \cos^2 (\alpha + \theta) = \frac{3}{2}$, where $\alpha = 60^\circ$ or 120° or 240° or 300° .
- (58) $\cos^3 \theta + \cos^3 (120^\circ - \theta) + \cos^3 (120^\circ + \theta) = \frac{3}{4} \cos 3\theta.$
- (59) $(1 + \sec 2\theta)(1 + \sec 4\theta)(1 + \sec 8\theta) \dots (1 + \sec 2^n \theta) = \tan 2^n \theta \cot \theta.$
- (60) $(2 \cos \theta - 1)(2 \cos 2\theta - 1)(2 \cos 4\theta - 1) \dots (2 \cos^{n-1} \theta - 1) = \frac{2 \cos 2^n \theta + 1}{2 \cos \theta + 1}.$

• **Transformation:**

- (1) $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}.$
- (2) $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}.$
- (3) $\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}.$
- (4) $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2} = \sin \frac{C+D}{2} \sin \frac{D-C}{2}.$
- (5) $2 \sin A \cos B = \sin(A+B) + \sin(A-B).$
- (6) $2 \cos A \sin B = \sin(A+B) - \sin(A-B).$
- (7) $2 \cos A \cos B = \cos(A+B) + \cos(A-B).$
- (8) $2 \sin A \sin B = \cos(A-B) - \cos(A+B).$
- (9) $\tan \frac{A+B}{2} = \frac{\sin A + \sin B}{\cos A + \cos B}.$
- (10) $\tan \frac{A-B}{2} = \frac{\sin A - \sin B}{\cos B - \cos A}.$
- (11) $\sin(A+B) = \frac{2(\sin A + \sin B)(\cos A + \cos B)}{(\sin A + \sin B)^2 + (\cos A + \cos B)^2}.$
- (12) $\cos(A+B) = \frac{(\cos A + \cos B)^2 - (\sin A + \sin B)^2}{(\cos A + \cos B)^2 + (\sin A + \sin B)^2}.$

- (13) $\tan(A+B) = \frac{2(\sin A + \sin B)(\cos A + \cos B)}{(\cos A + \cos B)^2 - (\sin A + \sin B)^2}.$
- (14) If in a triangle ABC, $\cos^2 A + \cos^2 B + \cos^2 C = 1$ or $\sin^2 A + \sin^2 B + \sin^2 C = 2$ then the triangle is right angled triangle.
- (15) If the angles A, B, C of a triangle ABC are in AP then $B = \frac{A+C}{2}$ and $B = 60^\circ$.
- (16) If $\sin(y+z-x)$, $\sin(z+x-y)$, $\sin(x+y-z)$ are in AP, then $\tan x, \tan y, \tan z$ are in AP.
- If $A+B+C = 180^\circ$, then:
 - (1) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C.$
 - (2) $\sin 2A + \sin 2B - \sin 2C = 4 \cos A \cos B \sin C.$
 - (3) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C.$
 - (4) $\cos 2A + \cos 2B - \cos 2C = 1 - 4 \sin A \sin B \cos C.$
 - (5) $\sin A + \sin B + \sin C = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$
 - (6) $\sin A + \sin B - \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}.$
 - (7) $\cos A + \cos B + \cos C = -1 - 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}.$
 - (8) $\cos A + \cos B - \cos C = 1 - 4 \sin \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}.$
 - (9) $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2 \cos A \cos B \cos C.$
 - (10) $\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C.$
 - (11) $\cos^2 A + \cos^2 B + \cos^2 C = 1 - 2 \cos A \cos B \cos C.$
 - (12) $\cos^2 A + \cos^2 B - \cos^2 C = 1 - 2 \sin A \sin B \cos C.$
 - (13) $\cos^2 \frac{A}{2} + \cos^2 \frac{B}{2} + \cos^2 \frac{C}{2} = 1 - 2 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$
 - (14) $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} = 2 + 2 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$
 - (15) $\tan A + \tan B + \tan C = \tan A \tan B \tan C.$
 - (16) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1.$
 - (17) $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1.$
 - (18) $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} = \cot \frac{A}{2} \cot \frac{B}{2} \cot \frac{C}{2}.$
 - **Periodicity:** A real function $f : A \rightarrow B$ is such that $f(x+t) = f(x)$, $\forall t \in R$, then f is called as periodic function and the least positive number t is called as period of the function f .
 - (1) If period of $f(x)$ is t then period of $-f(x)$ is also t .
 - (2) If period of $f(x)$ is t then period of $f(ax)$ is $\frac{t}{a}$.
 - (3) If period of $f(x)$ is t then period of $f\left(\frac{x}{a}\right)$ is at .
 - (4) The period of $f(x) = \sin x$ is 2π .
 - (5) The period of $f(x) = \cos x$ is 2π .

- (6) The period of $f(x) = \operatorname{cosec} x$ is 2π .
- (7) The period of $f(x) = \sec x$ is 2π .
- (8) The period of $f(x) = \tan x$ is π .
- (9) The period of $f(x) = \cot x$ is π .
- (10) The period of $f(x) = \sin kx$ is $\frac{2\pi}{|k|}$.
- (11) The period of $f(x) = \cos kx$ is $\frac{2\pi}{|k|}$.
- (12) The period of $f(x) = \operatorname{cosec} kx$ is $\frac{2\pi}{|k|}$.
- (13) The period of $f(x) = \sec kx$ is $\frac{2\pi}{|k|}$.
- (14) The period of $f(x) = \tan kx$ is $\frac{\pi}{|k|}$.
- (15) The period of $f(x) = \cot kx$ is $\frac{\pi}{|k|}$.
- (16) The period of $f(x) = \sin^n x$ is $\begin{cases} \pi & n = \text{even} \\ 2\pi & n = \text{odd} \end{cases}$.
- (17) The period of $f(x) = \cos^n x$ is $\begin{cases} \pi & n = \text{even} \\ 2\pi & n = \text{odd} \end{cases}$.
- (18) The period of $f(x) = \operatorname{cosec}^n x$ is $\begin{cases} \pi & n = \text{even} \\ 2\pi & n = \text{odd} \end{cases}$.
- (19) The period of $f(x) = \sec^n x$ is $\begin{cases} \pi & n = \text{even} \\ 2\pi & n = \text{odd} \end{cases}$.
- (20) The period of $f(x) = \tan^n x$ is π .
- (21) The period of $f(x) = \cot^n x$ is π .
- (22) The period of $f(x) = |\sin x|$ is π .
- (23) The period of $f(x) = |\cos x|$ is π .
- (24) The period of $f(x) = |\operatorname{cosec} x|$ is π .
- (25) The period of $f(x) = |\sec x|$ is π .
- (26) The period of $f(x) = |\tan x|$ is π .
- (27) The period of $f(x) = |\cot x|$ is π .

- (28) The period of $f(x) = a \sin^n x + b \cos^n x$ is

	$a = b$	$a \neq b$
$n \text{ even}$	$\frac{\pi}{2}$	π
$n \text{ odd}$	2π	2π

	$a = b$	$a \neq b$
(29) The period of $f(x) = a \tan^n x + b \cot^n x$ is	$\frac{\pi}{2}$	π
	2π	2π

	$a = b$	$a \neq b$
(30) The period of $f(x) = a \tan^n x + b \cot^n x$ is	$\frac{\pi}{2}$	π
	2π	2π

(31) The period of $f(x) = a|\sin x| + b|\cos x|$ is $\begin{cases} \frac{\pi}{2} & a = b \\ \pi & a \neq b \end{cases}$.

(32) The period of $f(x) = a|\tan x| + b|\cot x|$ is $\begin{cases} \frac{\pi}{2} & a = b \\ \pi & a \neq b \end{cases}$.

(33) The period of $f(x) = a|\operatorname{cosec} x| + b|\sec x|$ is $\begin{cases} \frac{\pi}{2} & a = b \\ \pi & a \neq b \end{cases}$.

(34) The period of $f(x) = x - [x]$ is 1.

(35) The period of $f(x) = ax - [ax]$ is $\frac{1}{a}$.

(36) If period of $f(x)$ is t_1 and period of $g(x)$ is t_2 then period of $af(x) \pm bg(x)$ is LCM of (t_1, t_2) where $a \neq b$.

(37) The maximum value of $a \cos x + b \sin x + c$ is $c + \sqrt{a^2 + b^2}$.

(38) The minimum value of $a \cos x + b \sin x + c$ is $c - \sqrt{a^2 + b^2}$.

(39) The range of $a \cos x + b \sin x + c$ is $[c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}]$.

(40) The range of $a^2 \sin^2 x + b^2 \operatorname{cosec}^2 x$ is $[2ab, \infty)$.

(41) The range of $a^2 \cos^2 x + b^2 \sec^2 x$ is $[2ab, \infty)$.

(42) The range of $a^2 \tan^2 x + b^2 \cot^2 x$ is $[2ab, \infty)$.

(43) The range of $a^2 \sec^2 x + b^2 \operatorname{cosec}^2 x$ is $[(a+b)^2, \infty)$.

(44) The minimum value of $a^2 \sin^2 x + b^2 \operatorname{cosec}^2 x$ is $2ab$.

(45) The minimum value of $a^2 \cos^2 x + b^2 \sec^2 x$ is $2ab$.

(46) The minimum value of $a^2 \tan^2 x + b^2 \cot^2 x$ is $2ab$.

(47) The minimum value of $a^2 \sec^2 x + b^2 \operatorname{cosec}^2 x$ is $(a+b)^2$ at $x = \tan^{-1} \left(\sqrt{\frac{b}{a}} \right)$.

(48) The minimum value of $a \sin^2 x + b \sin x \cos x + c \cos^2 x$ is $\frac{a+c - \sqrt{b^2 + (a-c)^2}}{2}$.

(49) The maximum value of $a \sin^2 x + b \sin x \cos x + c \cos^2 x$ is $\frac{a+c + \sqrt{b^2 + (a-c)^2}}{2}$.

(50) The range of $a \sin^2 x + b \sin x \cos x + c \cos^2 x$ is:

$$\left[\frac{a+c-\sqrt{b^2+(a-c)^2}}{2}, \frac{a+c+\sqrt{b^2+(a-c)^2}}{2} \right].$$

- **Trigonometric Equations:** The equation involving trigonometric functions of unknown angles are called as trigonometric equations.
- **Solution of Trigonometric Equations:** A solution of a trigonometric equation is the value of the unknown angle that satisfies the equation. The trigonometric equations may have infinite number of solutions.

- (1) **Principal Solution:** Numerical least angle within the domain is called as principal solution.
- (2) **General Solution:** The solution consisting of possible solutions of trigonometric equations is called as its general solution.

- If $\sin \theta = 0$ then general solution is $\theta = n\pi$ where $n \in Z$.
- If $\sin \theta = 1$ then general solution is $\theta = (2n+1)\frac{\pi}{2}$ where $n \in Z$.
- If $\sin \theta = -1$ then general solution is $\theta = (4n-1)\frac{\pi}{2}$ where $n \in Z$.
- If $\cos \theta = 0$ then general solution is $\theta = (2n+1)\frac{\pi}{2}$ where $n \in Z$.
- If $\cos \theta = 1$ then general solution is $\theta = 2n\pi$ where $n \in Z$.
- If $\cos \theta = -1$ then general solution is $\theta = (2n+1)\pi$ where $n \in Z$.
- If $\tan \theta = 0$ then general solution is $\theta = n\pi$ where $n \in Z$.
- If $\sin \theta = \sin \alpha$ then general solution is $\theta = n\pi + (-1)^n \alpha$ where $n \in Z$.
- If $\cos \theta = \cos \alpha$ then general solution is $\theta = 2n\pi \pm \alpha$ where $n \in Z$.
- If $\tan \theta = \tan \alpha$ then general solution is $\theta = n\pi + \alpha$ where $n \in Z$.
- If $\sin^2 \theta = \sin^2 \alpha$ then general solution is $\theta = n\pi \pm \alpha$ where $n \in Z$.
- If $\cos^2 \theta = \cos^2 \alpha$ then general solution is $\theta = n\pi \pm \alpha$ where $n \in Z$.
- If $\tan^2 \theta = \tan^2 \alpha$ then general solution is $\theta = n\pi \pm \alpha$ where $n \in Z$.
- If $a \cos \theta + b \sin \theta = c$ then general solution is $\theta = 2n\pi \pm \beta + \alpha$ where $\alpha = \tan^{-1}\left(\frac{b}{a}\right)$ and

$$\beta = \cos^{-1}\left(\frac{c}{\sqrt{a^2+b^2}}\right).$$

- **Tips:**
 - (1) $\sin n\pi = 0$.
 - (2) $\cos n\pi = (-1)^n$.
 - (3) $\cos 2n\pi = 1$.
 - (4) $\sin(n\pi \pm \theta) = \pm(-1)^n \sin \theta$.
 - (5) $\cos(n\pi \pm \theta) = (-1)^n \cos \theta$.

- (6) $\tan(n\pi \pm \theta) = \pm \tan \theta$.
- (7) $\sin\left(\frac{n\pi}{2} + \theta\right) = \begin{cases} (-1)^{\frac{n-1}{2}} \cos \theta & n \rightarrow \text{odd} \\ (-1)^{\frac{n}{2}} \sin \theta & n \rightarrow \text{even} \end{cases}$.
- (8) $\cos\left(\frac{n\pi}{2} + \theta\right) = \begin{cases} (-1)^{\frac{n+1}{2}} \sin \theta & n \rightarrow \text{odd} \\ (-1)^{\frac{n}{2}} \cos \theta & n \rightarrow \text{even} \end{cases}$.
- (9) $\tan\left(\frac{A}{2}\right) = \frac{-1 \pm \sqrt{1 + \tan^2 A}}{\tan A}$.
- (10) $\tan \alpha = \cot \alpha - 2 \cot 2\alpha$.
- (11) $\operatorname{cosec} 2\alpha = \cot \alpha - \cot 2\alpha$.
- (12) $\sec \alpha \sec(\alpha + \beta) = \operatorname{cosec} \beta \{\tan(\alpha + \beta) - \tan \alpha\}$.
- (13) $\tan \alpha \tan(\alpha + \beta) = \cot \beta \{\tan(\alpha + \beta) - \tan \alpha\} - 1$.
- (14) $\operatorname{cosec} \alpha \operatorname{cosec}(\alpha + \beta) = \operatorname{cosec} \beta \{\cot \alpha - \cot(\alpha + \beta)\}$.
- (15) $\tan \alpha \sec 2\alpha = \tan 2\alpha - \tan \alpha$.
- (16) $\tan^2 \alpha \tan 2\alpha = \tan 2\alpha - 2 \tan \alpha$.
- (17) $1 + \tan A \tan B = \frac{\cos(A - B)}{\cos A \cos B}$.
- (18) $1 - \tan A \tan B = \frac{\cos(A + B)}{\cos A \cos B}$.
- (19) $1 + \cot A \cot B = \frac{\cos(A - B)}{\sin A \sin B}$.
- (20) $1 - \cot A \cot B = \frac{-\cos(A + B)}{\sin A \sin B}$.
- (21) If $A + B + C = 0$, then $\sin A + \sin B + \sin C = -4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.
- (22) If $A + B + C = 0$, then $\cos A + \cos B + \cos C = -1 + 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$.
- (23) If $A + B + C = 0$, then $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.
- (24) $\tan A + \cot A = 2 \operatorname{cosec} 2A$.
- (25) $\cot A - \tan A = 2 \cot 2A$.
- (26) $\cos A + \cos B + \cos C + \cos(A + B + C) = 4 \cos \frac{A+B}{2} \cos \frac{B+C}{2} \cos \frac{C+A}{2}$.
- (27) $\sin A + \sin B + \sin C - \sin(A + B + C) = 4 \sin \frac{A+B}{2} \sin \frac{B+C}{2} \sin \frac{C+A}{2}$.
- (28) $\tan A + \tan B + \tan C - \tan(A + B + C) = 4 \tan A \tan B \tan C \left(1 - \frac{\cot A + \cot B + \cot C}{\cot(A + B + C)}\right)$.
- (29) $\sin A \sin(B - C) + \sin B \sin(C - A) + \sin C \sin(A - B) = 0$.
- (30) $\cos A \cos(B - C) + \cos B \cos(C - A) + \cos C \cos(A - B) = 0$.
- (31) $(\cos A + \sin A)(\cos B + \sin B) = \cos(A - B) + \sin(A + B)$.
- (32) $(\cos A - \sin A)(\cos B - \sin B) = \cos(A - B) - \sin(A + B)$.

Inverse Trigonometric Functions:

- $\sin(\sin^{-1} x) = x$ for $x \in [-1, 1]$.
- $\cos(\cos^{-1} x) = x$ for $x \in [-1, 1]$.
- $\tan(\tan^{-1} x) = x$ for $x \in (-\infty, \infty)$.
- $\cot(\cot^{-1} x) = x$ for $x \in (-\infty, \infty)$.
- $\sec(\sec^{-1} x) = x$ for $x \in (-\infty, -1] \cup [1, \infty)$.
- $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$ for $x \in (-\infty, -1] \cup [1, \infty)$.

- $\sin^{-1}(\sin x) = x$ for $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

- $\sin^{-1}(\sin x) = \begin{cases} x & 0 \leq x \leq \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x \leq \frac{3\pi}{2} \\ x - 2\pi & \frac{3\pi}{2} < x \leq 2\pi \end{cases}$.

- $\cos^{-1}(\cos x) = x$ for $x \in [0, \pi]$.

- $\cos^{-1}(\cos x) = \begin{cases} x & 0 \leq x \leq \pi \\ 2\pi - x & \pi < x \leq 2\pi \end{cases}$.

- $\tan^{-1}(\tan x) = x$ for $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

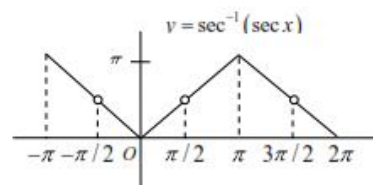
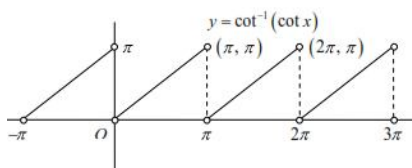
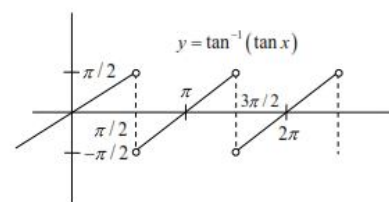
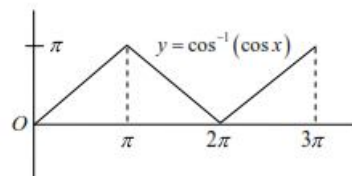
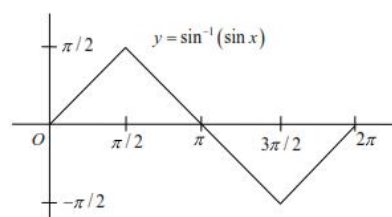
- $\tan^{-1}(\tan x) = \begin{cases} x & 0 \leq x < \frac{\pi}{2} \\ x - \pi & \frac{\pi}{2} < x < \frac{3\pi}{2} \\ x - 2\pi & \frac{3\pi}{2} < x \leq 2\pi \end{cases}$.

- $\cot^{-1}(\cot x) = x$ for $x \in (0, \pi)$.

- $\cot^{-1}(\cot x) = \begin{cases} x + \pi & -\pi < x < 0 \\ x & 0 < x < \pi \\ x - \pi & \pi < x < 2\pi \end{cases}$.

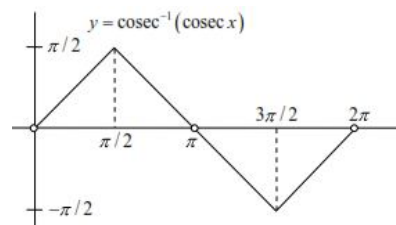
- $\sec^{-1}(\sec x) = x$ for $x \in \left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$.

- $\sec^{-1}(\sec x) = \begin{cases} -x & -\pi \leq x \leq 0, x \neq -\frac{\pi}{2} \\ x & 0 < x \leq \pi, x \neq \frac{\pi}{2} \end{cases}$.



- $\operatorname{cosec}^{-1}(\operatorname{cosec} x) = x$ for $x \in \left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$.

- $$\operatorname{cosec}^{-1}(\operatorname{cosec} x) = \begin{cases} x & 0 < x \leq \frac{\pi}{2} \\ \pi - x & \frac{\pi}{2} < x \leq \frac{3\pi}{2}, x \neq \pi \\ x - 2\pi & \frac{3\pi}{2} < x < 2\pi \end{cases}$$



- $\sin^{-1}(-x) = -\sin^{-1} x$ for $x \in [-1, 1]$.
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$ for $x \in [-1, 1]$.
- $\tan^{-1}(-x) = -\tan^{-1} x$ for $x \in (-\infty, \infty)$.
- $\cot^{-1}(-x) = \pi - \cot^{-1} x$ for $x \in (-\infty, \infty)$.
- $\sec^{-1}(-x) = \pi - \sec^{-1} x$ for $x \in (-\infty, -1] \cup [1, \infty)$.
- $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$ for $x \in (-\infty, -1] \cup [1, \infty)$.
- $\sin^{-1}(x) = \operatorname{cosec}^{-1}\left(\frac{1}{x}\right)$ for $x \in [-1, 0) \cup (0, 1]$.
- $\cos^{-1}(x) = \sec^{-1}\left(\frac{1}{x}\right)$ for $x \in [-1, 0) \cup (0, 1]$.
- $\tan^{-1}(x) = \cot^{-1}\left(\frac{1}{x}\right)$ for $x \in (0, \infty)$ and $\tan^{-1}(x) = -\pi + \cot^{-1}\left(\frac{1}{x}\right)$ for $x \in (-\infty, 0)$.
- $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right)$ for $x \in (0, \infty)$ and $\cot^{-1}(x) = \pi + \tan^{-1}\left(\frac{1}{x}\right)$ for $x \in (-\infty, 0)$.
- $\sec^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$ for $x \in (-\infty, -1] \cup [1, \infty)$.
- $\operatorname{cosec}^{-1}(x) = \sin^{-1}\left(\frac{1}{x}\right)$ for $x \in (-\infty, -1] \cup [1, \infty)$.
- $\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}$ for $x \in [-1, 1]$.
- $\tan^{-1}(x) + \cot^{-1}(x) = \frac{\pi}{2}$ for $x \in (-\infty, \infty)$.
- $\sec^{-1}(x) + \operatorname{cosec}^{-1}(x) = \frac{\pi}{2}$ for $x \in (-\infty, -1] \cup [1, \infty)$.
- $\sin^{-1}(x) = \cos^{-1}(\sqrt{1-x^2})$ for $x \in [0, 1]$.
- $\sin^{-1}(x) = -\cos^{-1}(\sqrt{1-x^2})$ for $x \in [-1, 0]$.
- $\sin^{-1}(x) = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ for $x \in (-1, 1)$.

- $\sin^{-1}(x) = \cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ for $x \in (0, 1]$.
- $\sin^{-1}(x) = \cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) - \pi$ for $x \in [-1, 0)$.
- $\cos^{-1}(x) = \sin^{-1}\left(\sqrt{1-x^2}\right)$ for $x \in [0, 1]$.
- $\cos^{-1}(x) = \pi - \sin^{-1}\left(\sqrt{1-x^2}\right)$ for $x \in [-1, 0]$.
- $\cos^{-1}(x) = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ for $x \in (0, 1]$.
- $\cos^{-1}(x) = \pi + \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right)$ for $x \in [-1, 0)$.
- $\cos^{-1}(x) = \cot^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ for $x \in (-1, 1)$.
- $\tan^{-1}(x) = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ for $x \geq 0$.
- $\tan^{-1}(x) = -\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ for $x \leq 0$.
- $\tan^{-1}(x) = \cot^{-1}\left(\frac{1}{x}\right)$ for $x > 0$.
- $\tan^{-1}(x) = \cot^{-1}\left(\frac{1}{x}\right) - \pi$ for $x < 0$.
- $\tan^{-1}(x) = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ for $x \in R$.
- $\cot^{-1}(x) = \sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ for $x \geq 0$.
- $\cot^{-1}(x) = \pi - \sin^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ for $x < 0$.
- $\cot^{-1}(x) = \tan^{-1}\left(\frac{1}{x}\right)$ for $x > 0$.
- $\cot^{-1}(x) = \pi + \tan^{-1}\left(\frac{1}{x}\right)$ for $x < 0$.
- $\cot^{-1}(x) = \cos^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$ for $x \in R$.

- If $x \in [0, 1]$ and $y \in [0, 1]$

- (1) $\sin^{-1}(x) + \sin^{-1}(y) = \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$ for $x^2 + y^2 \leq 1$.
- (2) $\sin^{-1}(x) + \sin^{-1}(y) = \pi - \sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$ for $x^2 + y^2 > 1$.
- (3) $\sin^{-1}(x) - \sin^{-1}(y) = \sin^{-1}(x\sqrt{1-y^2} - y\sqrt{1-x^2})$.
- (4) $\cos^{-1}(x) + \cos^{-1}(y) = \cos^{-1}(xy - \sqrt{1-x^2}\sqrt{1-y^2})$ for $x^2 + y^2 \leq 1$.
- (5) $\cos^{-1}(x) + \cos^{-1}(y) = \pi - \cos^{-1}(\sqrt{1-x^2}\sqrt{1-y^2} - xy)$ for $x^2 + y^2 > 1$.
- (6) $\cos^{-1}(x) - \cos^{-1}(y) = \cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$ for $x \leq y$.
- (7) $\cos^{-1}(x) - \cos^{-1}(y) = -\cos^{-1}(xy + \sqrt{1-x^2}\sqrt{1-y^2})$ for $x \geq y$.

- For $x > 0, y > 0, \tan^{-1}(x) + \tan^{-1}(y) = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ if $xy < 1$.

- For $x > 0, y > 0, \tan^{-1}(x) + \tan^{-1}(y) = \frac{\pi}{2}$ if $xy = 1$.

- For $x > 0, y > 0, \tan^{-1}(x) + \tan^{-1}(y) = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ if $xy > 1$.

- If x and y are of same sign then $\tan^{-1}(x) - \tan^{-1}(y) = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$.

- $2\sin^{-1}(x) = \sin^{-1}(2x\sqrt{1-x^2})$ if $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$.

- $2\sin^{-1}(x) = \pi - \sin^{-1}(2x\sqrt{1-x^2})$ if $\frac{1}{\sqrt{2}} < x \leq 1$.

- $2\sin^{-1}(x) = -\pi + \sin^{-1}(2x\sqrt{1-x^2})$ if $-1 \leq x < -\frac{1}{\sqrt{2}}$.

- $2\sin^{-1}(x) = \cos^{-1}(1-2x^2)$.

- $3\sin^{-1}(x) = \sin^{-1}(3x-4x^3)$ if $-\frac{1}{2} \leq x \leq \frac{1}{2}$.

- $3\sin^{-1}(x) = \pi - \sin^{-1}(3x-4x^3)$ if $\frac{1}{2} < x \leq 1$.

- $3\sin^{-1}(x) = -\pi - \sin^{-1}(3x-4x^3)$ if $-1 \leq x < -\frac{1}{2}$.

- $2\cos^{-1}(x) = \cos^{-1}(1-2x^2)$ if $0 \leq x \leq 1$.

- $2\cos^{-1}(x) = 2\pi - \cos^{-1}(1-2x^2)$ if $-1 \leq x < 0$.

- $2\cos^{-1}(x) = \sin^{-1}(2x\sqrt{1-x^2})$.

- $3\cos^{-1}(x) = \cos^{-1}(4x^3-3x)$ if $\frac{1}{2} \leq x \leq 1$.

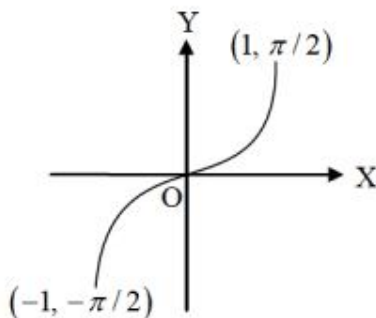
- $3\cos^{-1}(x) = 2\pi - \cos^{-1}(4x^3 - 3x)$ if $-\frac{1}{2} \leq x < \frac{1}{2}$.
- $3\cos^{-1}(x) = 2\pi + \cos^{-1}(4x^3 - 3x)$ if $-1 \leq x < -\frac{1}{2}$.
- $2\tan^{-1}(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ if $-1 < x < 1$.
- $2\tan^{-1}(x) = \pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ if $x > 1$.
- $2\tan^{-1}(x) = -\pi + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$ if $x < -1$.
- $2\tan^{-1}(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ if $-1 \leq x \leq 1$.
- $2\tan^{-1}(x) = \pi + \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ if $x > 1$.
- $2\tan^{-1}(x) = -\pi + \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ if $x < -1$.
- $2\tan^{-1}(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ if $x \geq 0$.
- $2\tan^{-1}(x) = -\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ if $x < 0$.
- $3\tan^{-1}(x) = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ if $-\frac{1}{\sqrt{3}} \leq x \leq \frac{1}{\sqrt{3}}$.
- $3\tan^{-1}(x) = \pi + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ if $x > \frac{1}{\sqrt{3}}$.
- $3\tan^{-1}(x) = -\pi + \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$ if $x < -\frac{1}{\sqrt{3}}$.
- $\tan^{-1}(x) + \tan^{-1}(y) + \tan^{-1}(z) = \tan^{-1}\left(\frac{x+y+z-xyz}{1-xy-yz-zx}\right)$.
- $\tan^{-1}(x) + \tan^{-1}(y) + \tan^{-1}(z) = \frac{\pi}{2}$ if $xy + yz + zx = 1$.
- $\tan^{-1}(x) + \tan^{-1}(y) + \tan^{-1}(z) = \pi$ if $x + y + z = xyz$.
- $\tan^{-1}\left(\frac{n}{m}\right) + \tan^{-1}\left(\frac{m-n}{m+n}\right) = \frac{\pi}{4}$ or $-\frac{3\pi}{4}$.
- $\tan^{-1}\left(\frac{m}{n}\right) - \tan^{-1}\left(\frac{m-n}{m+n}\right) = \frac{\pi}{4}$ or $-\frac{3\pi}{4}$.
- $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$.

- $\tan^{-1}(2) + \tan^{-1}(3) = \frac{3\pi}{4}$.
- $\tan^{-1}(1) + \tan^{-1}(2) + \tan^{-1}(3) = \pi$.
- If $\tan^{-1}(x) + \tan^{-1}(y) = \frac{\pi}{2}$, then $xy = 1$.
- If $\cot^{-1}(x) + \cot^{-1}(y) = \frac{\pi}{2}$, then $xy = 1$.
- If $\sin^{-1}\left(\frac{a}{x}\right) + \sin^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$ or $\cos^{-1}\left(\frac{a}{x}\right) + \cos^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$, then $x = \sqrt{a^2 + b^2}$.
- If $\tan^{-1}\left(\frac{a}{x}\right) + \tan^{-1}\left(\frac{b}{x}\right) = \frac{\pi}{2}$, then $x = \sqrt{ab}$.
- If $\cos^{-1}\left(\frac{x}{a}\right) + \cos^{-1}\left(\frac{y}{b}\right) = \theta$, then $\frac{x^2}{a^2} - \frac{2xy}{ab} \cos \theta + \frac{b^2}{y^2} = \sin^2 \theta$.
- If $\sin^{-1}\left(\frac{x}{a}\right) + \sin^{-1}\left(\frac{y}{b}\right) = \theta$, then $\frac{x^2}{a^2} + \frac{2xy}{ab} \cos \theta + \frac{b^2}{y^2} = \sin^2 \theta$.
- If $\tan^{-1} \frac{1}{1+x(x+1)} + \tan^{-1} \frac{1}{1+(x+1)(x+2)} + \dots + \tan^{-1} \frac{1}{1+(x+n-1)(x+n)} = \tan^{-1}(x+n) - \tan^{-1} x$.

Graphs of Inverse Trigonometric Functions:

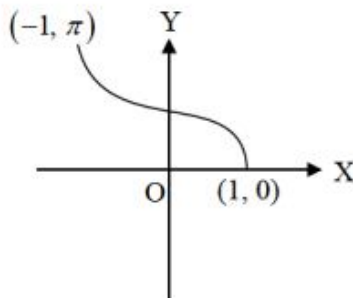
- **Graph of $y = \sin^{-1} x$.**

(i) Graph of $y = \sin^{-1} x$



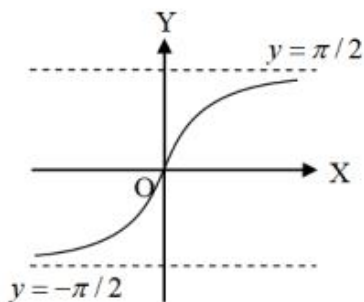
- **Graph of $y = \cos^{-1} x$.**

(ii) Graph of $y = \cos^{-1} x$



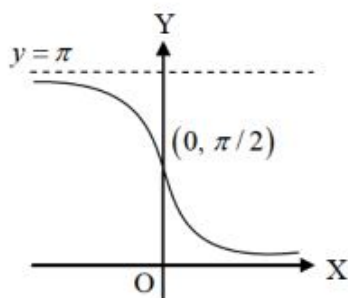
- Graph of $y = \tan^{-1} x$.

(iii) Graph of $y = \tan^{-1} x$



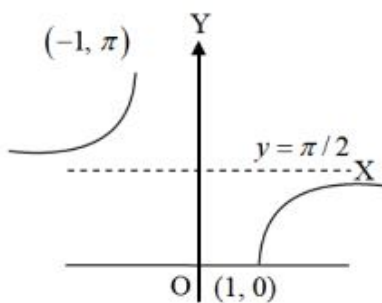
- Graph of $y = \cot^{-1} x$.

(iv) Graph of $y = \cot^{-1} x$



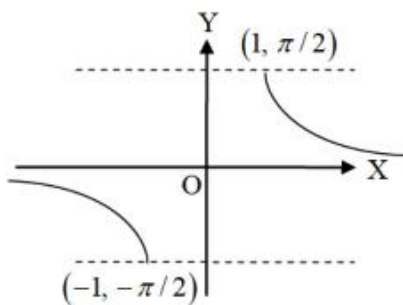
- Graph of $y = \sec^{-1} x$.

(v) Graph of $y = \sec^{-1} x$



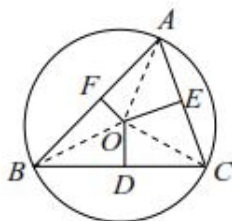
- Graph of $y = \operatorname{cosec}^{-1} x$.

(vi) Graph of $y = \operatorname{cosec}^{-1} x$



Properties & Solution of Triangle, Height & Distance:

- **Sine Rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, where a, b, c are sides opposite to angle A, B and C respectively, R is the radius of circumcircle.

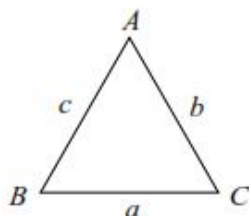


- **Cosine Rule:**

$$(1) \quad a^2 = b^2 + c^2 - 2bc \cos A \Rightarrow \cos A = \frac{b^2 + c^2 - a^2}{2bc}.$$

$$(2) \quad b^2 = c^2 + a^2 - 2ca \cos B \Rightarrow \cos B = \frac{c^2 + a^2 - b^2}{2ca}.$$

$$(3) \quad c^2 = a^2 + b^2 - 2ab \cos C \Rightarrow \cos C = \frac{a^2 + b^2 - c^2}{2ab}.$$



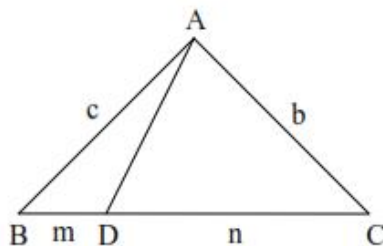
- **Projection Formula:**

$$(1) \quad a = b \cos C + c \cos B.$$

$$(2) \quad b = c \cos A + a \cos C.$$

$$(3) \quad c = a \cos B + b \cos C.$$

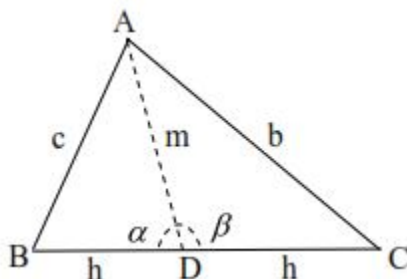
- **Apollonius Theorem:**



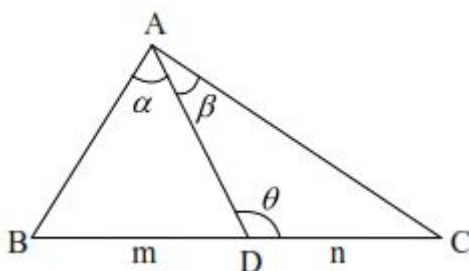
$$(1) \quad (m+n)AD^2 = mb^2 + nc^2 - mCD^2 - nBD^2.$$

$$(2) \quad (m+n)AD^2 = (m+n)(mb^2 + nc^2) - a^2 mn.$$

- **Apollonius Theorem for Medians:** In every triangle the sum of the squares of any two sides is equal to twice the square on half the third side together with twice the square on the medians that bisects the third side. $b^2 + c^2 = 2(h^2 + m^2) = 2\left(m^2 + \left(\frac{a}{2}\right)^2\right)$



- **Angle Between Median and the side of a triangle and The m-n Rule:**



- (1) $\sin \theta = \frac{2b}{\sqrt{2b^2 + 2c^2 - a^2}} \sin C$
- (2) $(m+n) \cot \theta = n \cot \alpha - n \cot B.$
- (3) $(m+n) \cot \theta = n \cot \beta - n \cot C.$

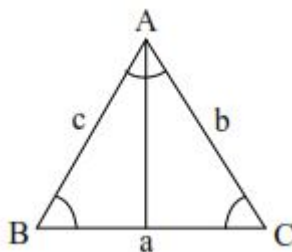
- **Napier's Analogy (Law of Tangents):** For any triangle ABC,

- (1) $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b} \cot \frac{C}{2}.$
- (2) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}.$
- (3) $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a} \cot \frac{B}{2}.$

- **Mollweide's Formula:**

- (1) $\frac{a+b}{c} = \frac{\cos \frac{1}{2}(A-B)}{\sin \frac{C}{2}}.$
- (2) $\frac{a-b}{c} = \frac{\sin \frac{1}{2}(A-B)}{\cos \frac{C}{2}}.$

• **Area of Triangle:**



(1) **When two sides and included angle given:**

$$\text{Area of triangle} = \Delta = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B = \frac{1}{2}ab \sin C.$$

$$\text{Area of triangle} = \Delta = \frac{1}{2}(\text{product of two sides}) \times \text{sine of the included angle}.$$

(2) **When three sides are given:**

$$\text{Area of triangle} = \Delta = \sqrt{s(s-a)(s-b)(s-c)} \text{ where } s = \frac{a+b+c}{2}.$$

(3) **When three sides and Circum-radius given:**

$$\text{Area of triangle} = \Delta = \frac{abc}{4R} \text{ where } R \text{ is the circum-radius.}$$

(4) **When two angles and included side given:**

$$\text{Area of triangle} = \Delta = \frac{1}{2}a^2 \frac{\sin B \sin C}{\sin(B+C)} = \frac{1}{2}b^2 \frac{\sin C \sin A}{\sin(C+A)} = \frac{1}{2}c^2 \frac{\sin A \sin B}{\sin(A+B)}.$$

- **Half Angle Formula:** a, b, c are sides and $s = \frac{a+b+c}{2}$ be the semi-perimeter of a triangle, then:

$$(1) \quad \sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}.$$

$$(2) \quad \sin \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}.$$

$$(3) \quad \sin \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}.$$

$$(4) \quad \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}.$$

$$(5) \quad \cos \frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}.$$

$$(6) \quad \cos \frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}.$$

$$(7) \quad \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{(s-b)(s-c)}{\Delta} \text{ where } \Delta \text{ be the area of triangle.}$$

$$(8) \quad \tan \frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{s(s-b)}} = \frac{(s-c)(s-a)}{\Delta} \text{ where } \Delta \text{ be the area of triangle.}$$

$$(9) \quad \tan \frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}} = \frac{(s-a)(s-b)}{\Delta} \text{ where } \Delta \text{ be the area of triangle.}$$

$$(10) \quad \cot \frac{A}{2} = \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} = \frac{s(s-a)}{\Delta} \text{ where } \Delta \text{ be the area of triangle.}$$

$$(11) \quad \cot \frac{B}{2} = \sqrt{\frac{s(s-b)}{(s-c)(s-a)}} = \frac{s(s-b)}{\Delta} \text{ where } \Delta \text{ be the area of triangle.}$$

$$(12) \quad \cot \frac{C}{2} = \sqrt{\frac{s(s-c)}{(s-a)(s-b)}} = \frac{s(s-c)}{\Delta} \text{ where } \Delta \text{ be the area of triangle.}$$

- **Oblique Triangle:** A triangle which is not right triangle is called as oblique triangle. We can solve the triangle if we know three of its parts atleast one of its side. Different Cases:

- (1) Three sides are given.
- (2) Two sides and included angle given.
- (3) Two sides and angle opposite to one of them.
- (4) One side and two angles given.

- **Solution of triangle if three sides are given:** Let ABC is a triangle in which all the three sides a, b, c are given. Then we have to determine three angles A, B, C.

- (1) **To find angle A.**

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

$$\Rightarrow \log \left(\tan \frac{A}{2} \right) = \frac{1}{2} (\log(s-b) + \log(s-c) - \log(s-a))$$

Easily we can find the value of $\log \left(\tan \frac{A}{2} \right)$ and hence the value of $\tan \frac{A}{2}$ and then we

can find the value of $\frac{A}{2}$ and hence we can find A.

- (2) **To find angle B.**

$$\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

$$\Rightarrow \log \left(\tan \frac{A}{2} \right) = \frac{1}{2} (\log(s-b) + \log(s-c) - \log(s-a))$$

Easily we can find the value of $\log \left(\tan \frac{A}{2} \right)$ and hence the value of $\tan \frac{A}{2}$ and then we

can find the value of $\frac{A}{2}$ and hence we can find A.

(3) **To find angle C.**

$$A + B + C = 180^\circ$$

$$\Rightarrow C = 180^\circ - A - B.$$

- **Solution of triangle if two sides and included angle given:** Let ABC is a triangle in which two sides $b, c (b > c)$ and angle included between them A is given. Then we have to determine two angles B, C and one side a .

(1) **To find angle B and C:**

$$\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c} \cot \frac{A}{2}$$

$$\Rightarrow \log \tan\left(\frac{B-C}{2}\right) = \log(b-c) - \log(b+c) + \log \cot \frac{A}{2}$$

Easily we can find the value of $\log \tan\left(\frac{B-C}{2}\right)$ and hence the value of $\tan\left(\frac{B-C}{2}\right)$

and then we can find the value of $\frac{B-C}{2}$ ----- $\rightarrow (1)$.

$$\frac{B+C}{2} = 90^\circ - \frac{A}{2} \text{ ----- } \rightarrow (2)$$

By solving we can find B and C.

Note: If $b, c (b < c)$, then use $\tan\left(\frac{C-B}{2}\right) = \frac{c-b}{c+b} \cot \frac{A}{2}$.

(2) **To find side a:**

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\Rightarrow a = \frac{b \sin A}{\sin B}$$

$$\Rightarrow \log a = \log b + \log \sin A - \log \sin B$$

Easily we can find the value of $\log a$ and hence we can find a .

- **Solution of triangle if one side and two angles given:** Let ABC is a triangle in which one side a and two angles B, C are given. Then we have to determine two sides b, c and one angle A.

(1) **To find angle A:**

$$A + B + C = 180^\circ \Rightarrow A = 180^\circ - B - C.$$

(2) **To find Side b :**

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\Rightarrow b = \frac{a \sin B}{\sin A}$$

$$\Rightarrow \log b = \log a + \log \sin B - \log \sin A$$

Easily we can find the value of $\log b$ and hence we can find b .

(3) **To find Side c :**

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

$$\Rightarrow c = \frac{a \sin C}{\sin A}$$

$$\Rightarrow \log c = \log a + \log \sin C - \log \sin A$$

Easily we can find the value of $\log c$ and hence we can find c .

- **Solution of triangle if two side and one angle opposite to one of them given:** Let ABC is a triangle in which two side a, b and one angle A opposite to side a given. Then we have to determine two sides c and two angles B, C .

(1) **To find Angle B:**

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\Rightarrow \sin B = \frac{\sin A}{ab} \text{-----} \rightarrow (1)$$

We can calculate value of $\sin B$ and hence we can calculate angle B .

(2) **To find Angle C:**

$$A + B + C = 180^\circ \Rightarrow C = 180^\circ - B - A.$$

(3) **To find Side c :**

$$\frac{a}{\sin A} = \frac{c}{\sin C}$$

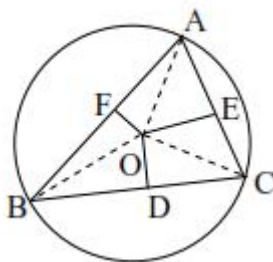
$$\Rightarrow c = \frac{a \sin C}{\sin A}$$

$$\Rightarrow \log c = \log a + \log \sin C - \log \sin A$$

Easily we can find the value of $\log c$ and hence we can find c .

Circle Connected with Triangles:

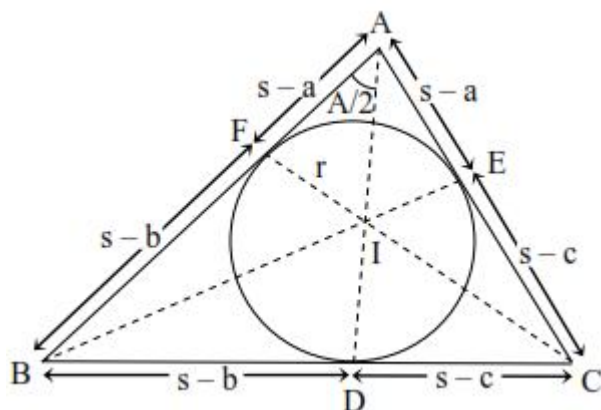
- **Circum-circle of a triangle and its radius:**



$$(1) \quad R = \frac{a}{2 \sin A} = \frac{b}{2 \sin B} = \frac{c}{2 \sin C}.$$

$$(2) \quad R = \frac{abc}{4\Delta}, \text{ where } \Delta \text{ is the area of triangle.}$$

• In-circle of a triangle and its radius:



(1) $r = \frac{\Delta}{s}$, where Δ is the area of triangle and s is the perimeter of triangle.

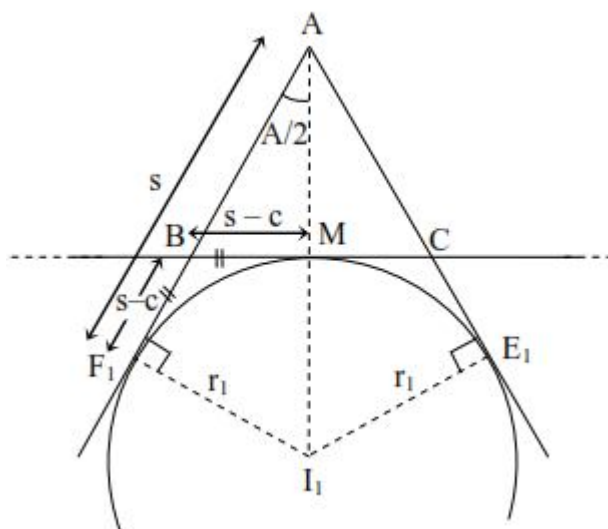
(2) $r = 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.

(3) $r = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2}$.

(4) $r = \frac{a \sin \frac{B}{2} \sin \frac{C}{2}}{\cos \frac{A}{2}} = \frac{b \sin \frac{C}{2} \sin \frac{A}{2}}{\cos \frac{B}{2}} = \frac{c \sin \frac{A}{2} \sin \frac{B}{2}}{\cos \frac{C}{2}}$.

(5) $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$.

• Ex-circle of a triangle and its radius:



$$(1) \quad r_1 = \frac{\Delta}{s-a} = \frac{a \cos \frac{B}{2} \cos \frac{C}{2}}{\cos \frac{A}{2}} = s \tan \frac{A}{2} = (s-c) \cot \frac{B}{2} = (s-c) \cot \frac{C}{2} = \frac{a}{\tan \frac{B}{2} + \tan \frac{C}{2}}.$$

$$(2) \quad r_2 = \frac{\Delta}{s-b} = \frac{b \cos \frac{C}{2} \cos \frac{A}{2}}{\cos \frac{B}{2}} = s \tan \frac{B}{2} = (s-c) \cot \frac{A}{2} = (s-a) \cot \frac{C}{2} = \frac{b}{\tan \frac{C}{2} + \tan \frac{A}{2}}.$$

$$(3) \quad r_3 = \frac{\Delta}{s-c} = \frac{c \cos \frac{A}{2} \cos \frac{B}{2}}{\cos \frac{C}{2}} = s \tan \frac{C}{2} = (s-a) \cot \frac{B}{2} = (s-b) \cot \frac{A}{2} = \frac{c}{\tan \frac{A}{2} + \tan \frac{B}{2}}.$$

$$(4) \quad r_1 + r_2 + r_3 - r = 4R.$$

$$(5) \quad \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}.$$

$$(6) \quad \frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{\Delta^2}.$$

$$(7) \quad \frac{1}{bc} + \frac{1}{ca} + \frac{1}{ab} = \frac{1}{2Rr}.$$

$$(8) \quad r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2.$$

$$(9) \quad \Delta = 2R^2 \sin A \sin B \sin C.$$

$$(10) \quad \Delta = 4Rr \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$$

$$(11) \quad r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$$

$$(12) \quad r_2 = 4R \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}.$$

$$(13) \quad r_3 = 4R \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}.$$

• **Distance of Circumcentre (O) from the Orthocentre(H), Incentre (I) and Excentre I_1, I_2, I_3 .**

(1) **Distance from Circumcentre (O) and Orthocentre (H):**

$$OH = R\sqrt{1 - 8\cos A \cos B \cos C}.$$

(2) **Distance from Circumcentre (O) and Incentre (I):**

$$OI = R\sqrt{1 - 8\sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}} = \sqrt{R(R - 2r)}.$$

- (3) **Distance from Circumcentre (O) and Excentre (I_1):**

$$OI_1 = R\sqrt{1 + 8\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}} = \sqrt{R}\sqrt{R + 2r_1}$$

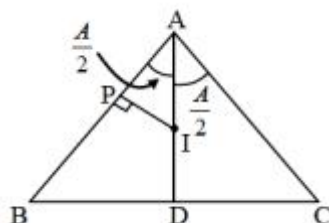
- (4) **Distance from Circumcentre (O) and Excentre (I_2):**

$$OI_2 = R\sqrt{1 + 8\cos\frac{A}{2}\sin\frac{B}{2}\cos\frac{C}{2}} = \sqrt{R}\sqrt{R + 2r_2}$$

- (5) **Distance from Circumcentre (O) and Excentre (I_3):**

$$OI_3 = R\sqrt{1 + 8\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}} = \sqrt{R}\sqrt{R + 2r_3}$$

- Distance of Incentre from the vertices of the Triangle:**



Let I be the incentre. Let $IP \perp AB$. Hence $IP = r$, $\angle PAI = \frac{A}{2}$.

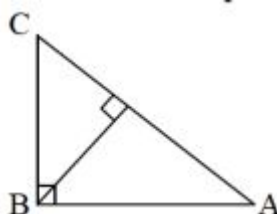
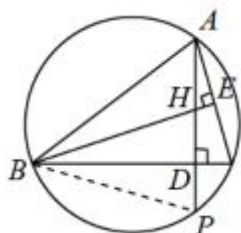
From right triangle IPA, $\sin\frac{A}{2} = \frac{r}{AI} \Rightarrow AI = r \operatorname{cosec}\frac{A}{2}$

$$(1) \quad AI = r \operatorname{cosec}\frac{A}{2}$$

$$(2) \quad BI = r \operatorname{cosec}\frac{B}{2}$$

$$(3) \quad CI = r \operatorname{cosec}\frac{C}{2}.$$

- Note Points:**

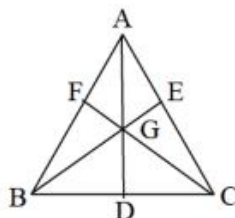


- The centroid of any triangle divides the join of circumcentre and orthocentre internally in the ratio 1:2.
- If H is the orthocentre of triangle ABC and AH produced meets BC at D and the circumcentre of triangle ABC at P, then $HD = DP$.

As $BD = c \cos B$ and $DP = BD \cot C$

So, $DP = c \cos B \cot C = 2R \cos B \cos C$.

- (3) The orthocentre of an acute angled triangle is the incentre of the pedal triangle.
- (4) The centre of circumcircle falls inside the triangle if triangle is acute angled but outside when it is obtuse angled. If the triangle is right angled triangle the centre lies on midpoint of the hypotenuse.
- (5) The orthocentre falls inside the triangle if triangle is acute angled and outside if triangle is obtuse angled triangle. If the triangle is right angled then orthocentre (B) lies on the triangle.



- (6) The length of median AD, BE and CF of triangle are given by:

- (a) $AD = \frac{1}{2} \sqrt{2b^2 + 2c^2 - a^2} = \frac{1}{2} \sqrt{b^2 + c^2 + 2bc \cos A}$.

- (b) $BE = \frac{1}{2} \sqrt{2c^2 + 2a^2 - b^2} = \frac{1}{2} \sqrt{c^2 + a^2 + 2ca \cos B}$.

- (c) $CF = \frac{1}{2} \sqrt{2a^2 + 2b^2 - c^2} = \frac{1}{2} \sqrt{a^2 + b^2 + 2ab \cos C}$.

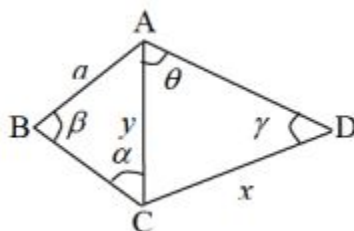
- (d) Distance between circum centre (O) and in-centre (I) of triangle ABC is $OI = R \sqrt{1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}}$.

- (e) Distance between circum centre (O) and ex-centre (I_1) opposite to angle A of triangle ABC is $OI_1 = R \sqrt{1 + 8 \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}}$.

- (f) Distance between circum centre (O) and ex-centre (I_2) opposite to angle B of triangle ABC is $OI_2 = R \sqrt{1 + 8 \cos \frac{A}{2} \sin \frac{B}{2} \cos \frac{C}{2}}$.

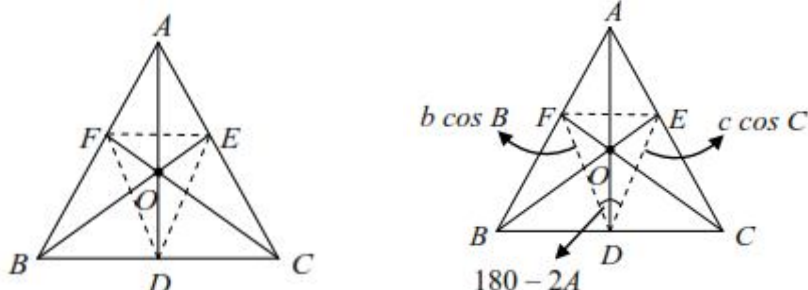
- (g) Distance between circum centre (O) and ex-centre (I_3) opposite to angle C of triangle ABC is $OI_3 = R \sqrt{1 + 8 \cos \frac{A}{2} \cos \frac{B}{2} \sin \frac{C}{2}}$.

- (7) **Application of Sine Rule:**



$$\begin{aligned}\frac{a}{\sin \alpha} &= \frac{y}{\sin \beta} \text{ and } \frac{x}{\sin \theta} = \frac{y}{\sin \gamma} \\ \Rightarrow \frac{x \sin \alpha}{a \sin \theta} &= \frac{\sin \beta}{\sin \gamma} \\ \Rightarrow x &= \frac{a \sin \theta \sin \beta}{\sin \alpha \sin \gamma}.\end{aligned}$$

• **Pedal Triangle:**



Let the perpendicular AD, BE and CF from the vertices A, B and C on the opposite sides BC, CA and AB of triangle ABC respectively meet at O. Then O is the orthocentre of triangle ABC. The triangle DEF is called as pedal triangle.

Orthocentre of the triangle is the incentre of the pedal triangle.

Let O, is the orthocentre of triangle ABC.

DEF is the Pedal triangle of triangle ABC.

- (1) $OA = 2R \cos A$.
- (2) $OB = 2R \cos B$.
- (3) $OC = 2R \cos C$.
- (4) $OD = 2R \cos B \cos C$.
- (5) $OE = 2R \cos C \cos A$.
- (6) $OF = 2R \cos A \cos B$.

(a) **Sides and Angles of a Pedal Triangle:**

If the triangle is acute angled:

- (1) **Angles of pedal triangle are:** $180^\circ - 2A$, $180^\circ - 2B$, $180^\circ - 2C$.
- (2) **Sides of pedal triangle are:** $EF = a \cos A = R \sin A$, $FD = b \cos B = R \sin B$ and $DE = c \cos C = R \sin C$.

If the triangle is obtuse angled:

- (1) **Angles of pedal triangle are:** $2A$, $2B$, $2C - 180^\circ$.
- (2) **Sides of pedal triangle are:** $EF = a \cos A = R \sin A$, $FD = b \cos B = R \sin B$ and $DE = -c \cos C = -R \sin C$.

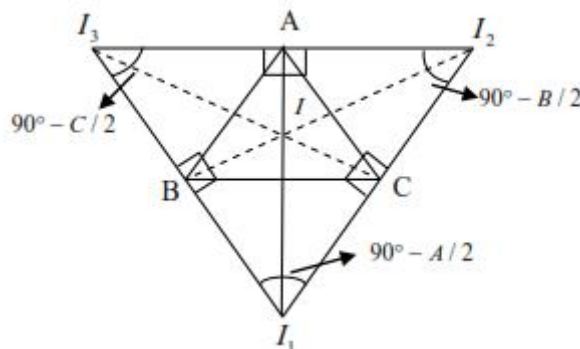
(b) **Area and Circum-radius and In-radius of Pedal Triangle:**

- (1) **Area of Pedal Triangle** $= \frac{1}{2} R^2 \sin 2A \sin 2B \sin 2C$
- (2) **Circum-radius** $= \frac{EF}{2 \sin(\angle FDE)} = \frac{R \sin 2A}{2 \sin(180^\circ - 2A)} = \frac{R}{2}$.

$$(3) \quad \text{In-radius} = \frac{\text{Area}(\triangle DEF)}{\text{Semi-perimeter } \triangle DEF} = \frac{\frac{1}{2} R^2 \sin 2A \sin 2B \sin 2C}{2R \sin A \sin B \sin C}.$$

$$\Rightarrow \text{In-radius} = 2R \cos A \cos B \cos C.$$

• **EX-Central Triangle:**



Let ABC be a triangle and I be the centre of in-circle. Let I_1, I_2, I_3 be the centre of escribed circle which are opposite to A, B, C respectively, then $I_1I_2I_3$ is called as Ex-Central triangle of triangle ABC.

If $I_1I_2I_3$ is a triangle, then triangle ABC be the pedal triangle of its ex-centra; triangle $I_1I_2I_3$.

(1) **Angles of ex-central triangle $I_1I_2I_3$** are: $90^\circ - \frac{A}{2}, 90^\circ - \frac{B}{2}, 90^\circ - \frac{C}{2}.$

(2) **Sides** $I_1I_2 = 4R \cos \frac{C}{2}, I_2I_3 = 4R \cos \frac{A}{2}, I_3I_1 = 4R \cos \frac{B}{2}.$

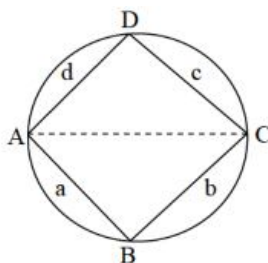
(3) **Length of** $I_1I = 4R \sin \frac{A}{2}, I_2I = 4R \sin \frac{B}{2}, I_3I = 4R \sin \frac{C}{2}.$

(4) **Area of Ex-central Triangle** $= \frac{1}{2} \left(4R \cos \frac{B}{2} \right) \left(4R \cos \frac{C}{2} \right) \sin \left(90^\circ - \frac{A}{2} \right)$

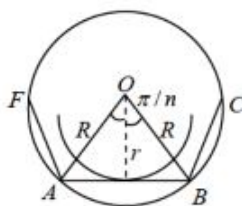
$$\Rightarrow \text{Area of Ex-central Triangle} = 8R^2 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}.$$

(5) **Circum-radius** $= \frac{I_2I_3}{2 \sin I_1I_2I_3} = \frac{4R \cos \frac{A}{2}}{2 \sin \left(90^\circ - \frac{A}{2} \right)} = 2R$

- **Cyclic Quadrilateral:** A quadrilateral ABCD is said to be cyclic quadrilateral if there exists a circle passing through all its four vertices A, B, C and D. Let a cyclic quadrilateral with $Ab = a, BC = b, CD = c, DA = d$. Let $2s = a + b + c + d$.



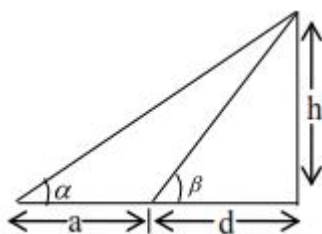
- (1) $\angle B + \angle D = 180^\circ$.
 - (2) $\angle A + \angle C = 180^\circ$.
 - (3) Area of cyclic quadrilateral $= \frac{1}{2}(ab + cd) \sin B$.
 - (4) Area of cyclic quadrilateral $= \sqrt{(s-a)(s-b)(s-c)(s-d)}$.
 - (5) $\cos B = \frac{a^2 + b^2 - c^2 - d^2}{2(ab + cd)}$
 - (6) $\cos D = \frac{c^2 + d^2 - a^2 - b^2}{2(ab + cd)}$.
 - (7) Circum-radius $= \frac{1}{4\Delta} \sqrt{(ac + bd)(ad + bc)(ab + cd)} = \frac{1}{4} \sqrt{\frac{(ac + bd)(ad + bc)(ab + cd)}{(s-a)(s-b)(s-c)(s-d)}}$.
- **Regular Polygon:** A regular polygon is a polygon which has all its sides equal and all angles are equal.



- (1) Sum of each interior angles of a regular polygon $= \pi(n-2)$.
- (2) Each interior angle of a regular polygon of n sides $= \frac{\pi(n-2)}{n}$.
- (3) Each exterior angles of a regular polygon of n sides $= \frac{2\pi}{n}$.
- (4) In-radius of the regular polygon $= r = \frac{1}{2}a \cot \frac{\pi}{n}$.
- (5) Circum-radius of the regular polygon $= R = \frac{1}{2}a \operatorname{cosec} \frac{\pi}{n}$.
- (6) Area of regular polygon $= n \times \text{Area of triangle OAB}$
 $\Rightarrow \text{Area of regular polygon} = \frac{1}{4}na^2 \cot \frac{\pi}{n}$ (in terms of side 'a').
 $\Rightarrow \text{Area of regular polygon} = nr^2 \tan \frac{\pi}{n}$ (in terms of in-radius 'r').
 $\Rightarrow \text{Area of regular polygon} = \frac{1}{2}nR^2 \sin \frac{2\pi}{n}$ (in terms of circum-radius 'R').

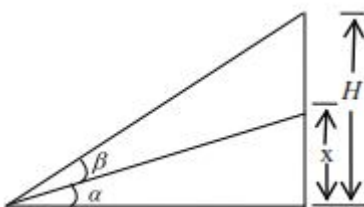
• Results of Height & Distance:

(1)



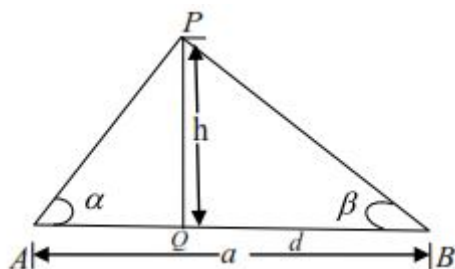
- (a) $a = h(\cot \alpha - \cot \beta) = \frac{h \sin(\beta - \alpha)}{\sin \alpha \sin \beta}.$
 (b) $h = a \sin \alpha \sin \beta \operatorname{cosec}(\beta - \alpha).$
 (c) $d = h \cot \beta = a \sin \alpha \cos \beta \operatorname{cosec}(\beta - \alpha)$

(2)



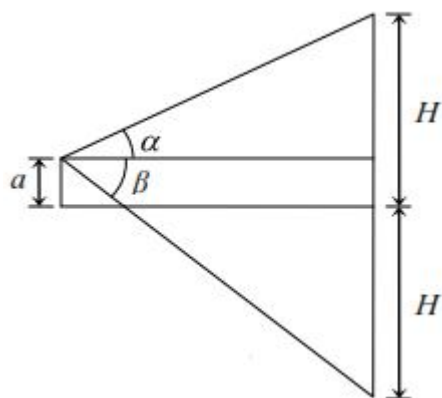
- (a) $H = x \cot \alpha + \tan(\alpha + \beta).$

(3)



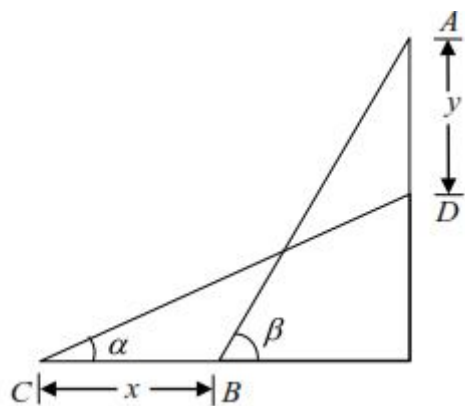
- (a) $a = h(\cot \alpha + \cot \beta).$
 (b) $h = a \sin \alpha \sin \beta \operatorname{cosec}(\alpha + \beta).$
 (c) $d = h \cot \beta = a \sin \alpha \cos \beta \operatorname{cosec}(\alpha + \beta)$

(4)



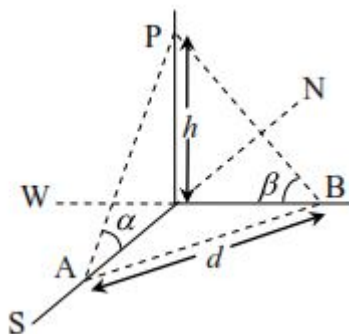
(a)
$$H = \frac{a \sin(\alpha + \beta)}{\sin(\beta - \alpha)}.$$

(5)



(a) If $B = CD$, then $x = y \tan \frac{\alpha + \beta}{2}.$

(6)



(a)
$$h = \frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta}}.$$

